LaTeX Author Guidelines for CVPR Proceedings

Anonymous CVPR submission

Paper ID *****

Abstract

[TODO: Abstract (150–200 words). Problem: floor-plan retrieval from incomplete queries (RPLAN). Method: frozen vision encoder + GNN on the room graph, combined by late fusion. Main result: fusion over the best single branch on the test split, read once (project status.md §53). Per-axis relevance only as descriptive analysis.]

1. Introduction

[TODO: Motivation: real design workflows start from partial layouts; retrieval must complete, not only resemble. Visual similarity differs from architectural relevance (function, topology). PAPER.md §2.]

[TODO: Why RPLAN: raster images and structural metadata for the same plans, so both an image branch and a graph branch are possible on the same gallery.]

[TODO: Our approach in one paragraph: vision branch + graph branch + late fusion, judged on robustness to incomplete queries.]

We make the following contributions:

- **[TODO: Contribution 1: two retrieval branches on the same gallery and queries, and their late fusion.]**
- **[TODO: Contribution 2: a robustness criterion over three damages that really remove information, fixed before the grid.]**
- **[TODO: Contribution 3: fusion beats the best branch under damage, with two controls that separate ensemble effect from complementary information.]**
- **[TODO: Contribution 4: negative results reported as such (rendering artifact, head that does not transfer).]**

2. Related Work

Floor-plan datasets. **[TODO: RPLAN; CubiCasa5K, ResPlan (out of scope, future work).]**

Graph representations and generation of floor plans. **[TODO: Graph2Plan and related graph-based layout models.]**

Floor-plan retrieval and similarity. **[TODO: LayoutGMN, LayoutGKN (graph similarity on RPLAN); other floor-plan CBIR.]**

Visual foundation encoders. **[TODO: The eight frozen encoders of the ablation; PE-spatial for the final config.]**

Graph neural networks and contrastive learning. **[TODO: GCN, GraphSAGE, GATv2; InfoNCE.]**

3. Method

[TODO: Overview: the two branches embed the same plans; fusion combines them; everything is evaluated on the same shared gallery (Fig. 2).]

3.1. Late fusion

[TODO: Weighted concatenation of the L2-normalised vectors $[\sqrt{\alpha}\, v;\ \sqrt{1-\alpha}\, g]$; α chosen on valid with pre-registered rules ($\alpha^* = 0.6$). Project status.md §49.]

3.2. Graph branch

[TODO: Room graph from the RPLAN metadata; GATv2 encoder trained with InfoNCE on asymmetric pairs (complete plan vs. plan with rooms removed); checkpoint selected on robustness. configs/graph_models/gat.yaml; project status.md §45–§47.]

3.3. Vision branch

[TODO: Frozen encoder (PE-spatial ViT-B/16) → GeM pooling → PCA whitening fitted on train → FAISS inner-product index. configs/vision_models/pespatial.yaml, configs/vision_retrieval.yaml.]

4. Experiments and Results

4.1. Metrics

Incomplete queries and robustness metric. **[TODO: Three damages at $f \in \{0.25, 0.5, 0.75\}$: rooms removed**

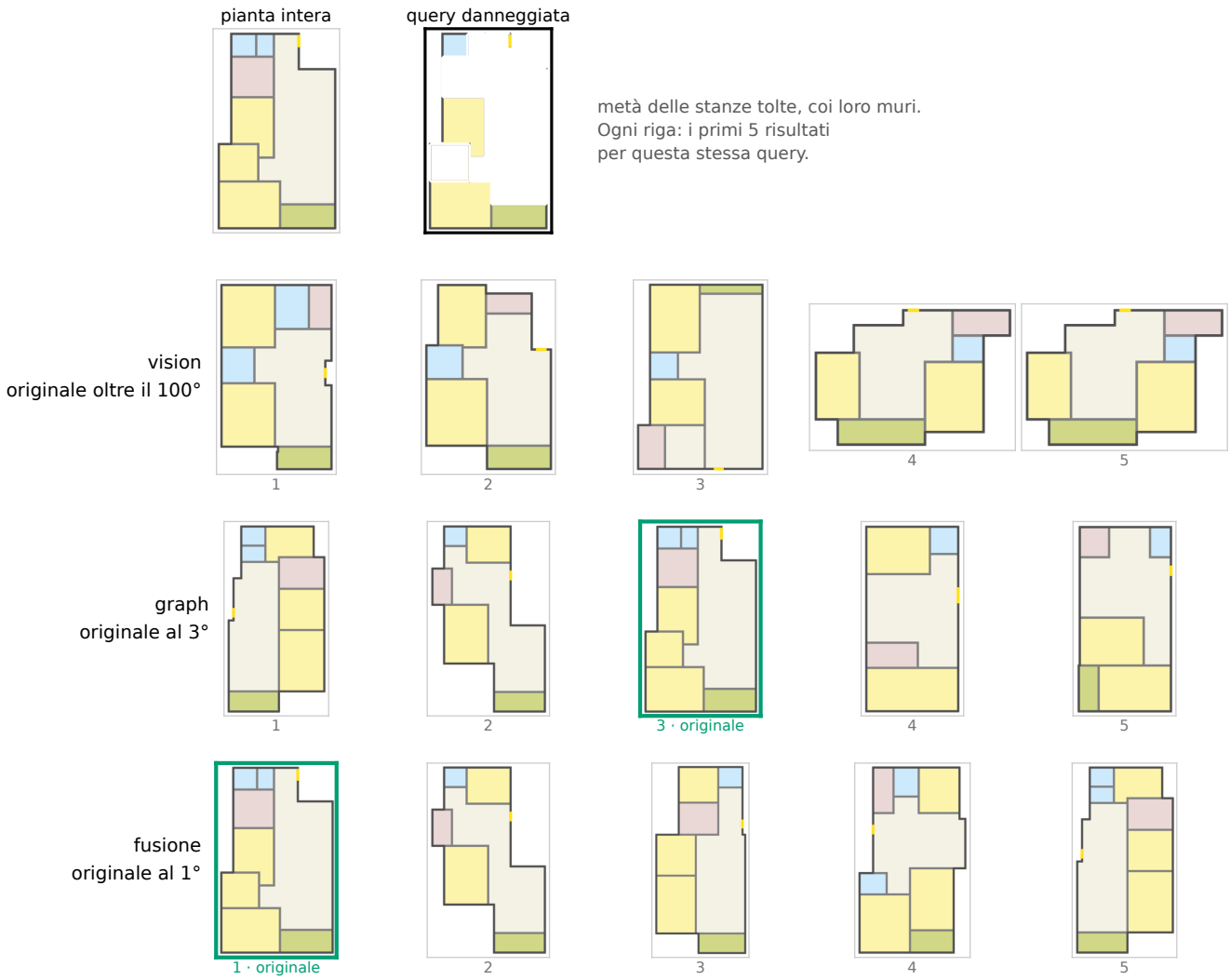


Figure 1. **[TODO: Teaser caption: one damaged query (half the rooms removed, valid split) and the top-5 results of vision, graph and fusion; the original plan is framed where it appears. State the conclusion. Source: f_teaser_valid.sources.txt.]**

with their walls, crop, patch. Self-recovery reciprocal rank → AUC over levels → R = mean of the three AUCs. Paired bootstrap 95% CI. PAPER.md §5.4; project status.md §38.]

Per-axis relevance (descriptive). **[TODO: Composition, topology, geometry; nDCG@10 on the full plan. Descriptive only: circular on every axis. PAPER.md §5.2–§5.3, §5.5.]**

4.2. Dataset and experimental details

[TODO: Table T1: RPLAN, official splits, shared gallery, 2000 queries (seed 42), valid selects, test read once, exclusions, random floor. Project status.md §20, §29.]

4.3. Results

4.3.1. Ablation

[TODO: Ablation table (mean ± std, pending D1): vision configurations under the three damages; graph; fusion weight α and the two controls. Main test result (T3, project status.md §53).]

Negative results. **[TODO: Wall artifact of the historical room masking (project status.md §35–§37); the trained projection head does not transfer to other damages (§37, §48); the lost-neighbours marker ties (§47).]**

4.3.2. Qualitative results

[TODO: Qualitative examples by model and on the dataset (Fig. 1).]

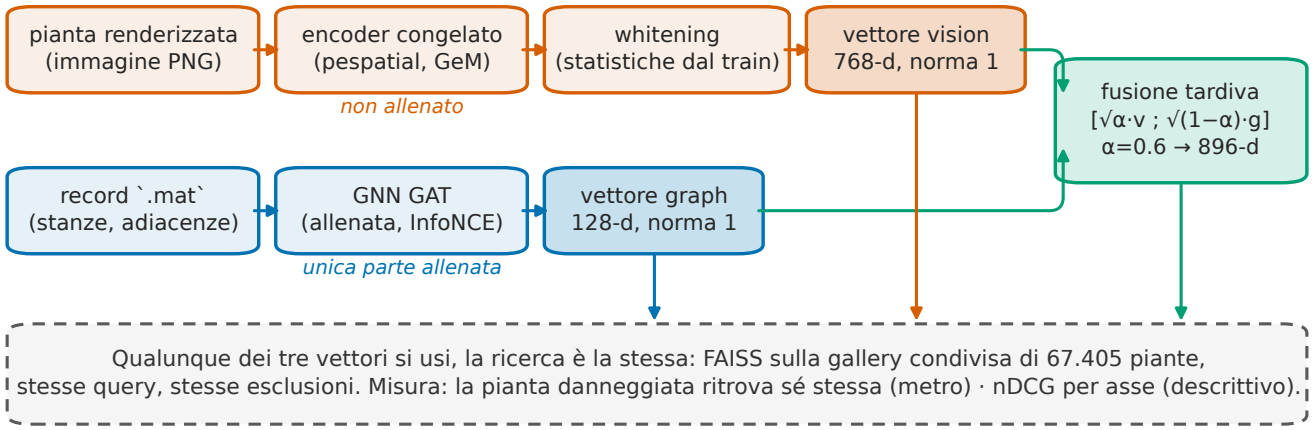


Figure 2. **[TODO: Pipeline caption: two branches on the same gallery, same queries, same measure; weighted concatenation with $\alpha = 0.6$. Source: f_pipeline.sources.txt.]**

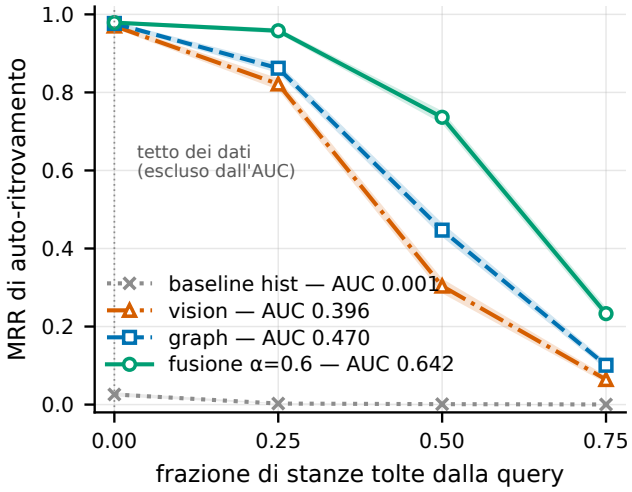


Figure 3. **[TODO: Test split: self-recovery vs. fraction of removed rooms for hist, vision, graph and fusion. Source: f_damage_test.sources.txt.]**

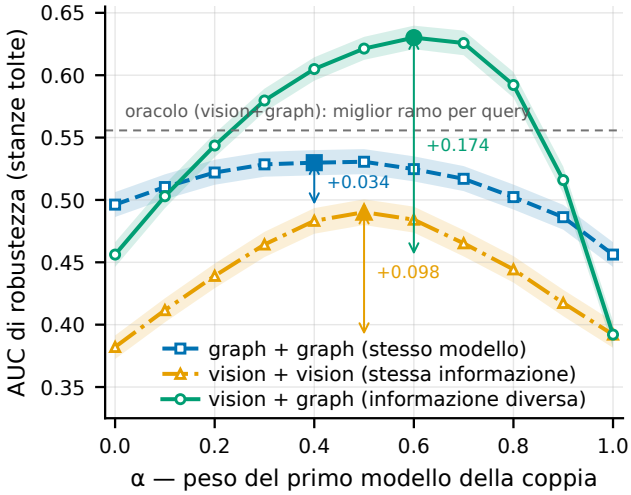


Figure 4. **[TODO: Valid split: robustness AUC vs. α for the three fused pairs. Source: f_alpha_valid.sources.txt.]**

4.4. Additional analysis: number of parameters

[TODO: Parameters per component: frozen vision backbone, trained GAT, fusion (none). Count on HPC with sum(p.numel()); not yet measured.]

5. Conclusions

[TODO: Summary; limitations (PAPER.md §8: RPLAN only, synthetic damages, proxy relevance, fusion tied to this graph checkpoint); future work (PAPER.md §9).]

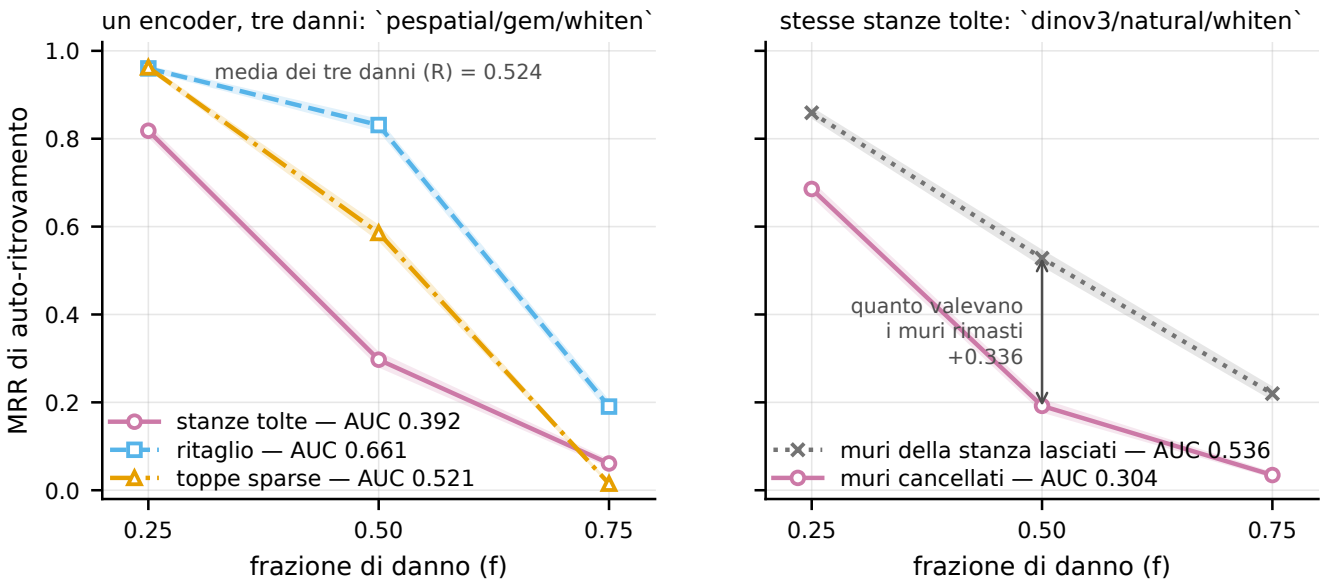


Figure 5. **[TODO: Left: final vision config under the three damages. Right: the wall artifact, same removed rooms with walls kept or erased. Source: f_damage_kinds_valid.sources.txt.]**